Fn-Logic:
The Mathematics of Process

A Fundamental Reorientation of Physical and Digital Computation

Author: Destiny Raiko Pulvermacher

E-Mail: Pulvermacher.Raiko@web.de

ORCID: 0009-0003-9431-1001

OSF: https://osf.io/py42t/

Working Paper / Preprint

DOI: https://doi.org/10.5281/zenodo.18757232

References:

Tensor Realities (TdR) -- DOI: https://doi.org/10.5281/zenodo.18727529

Energy Revolution -- DOI: https://doi.org/10.5281/zenodo.18778354

Abstract

Fn-Logic is presented as an additive process model that conceives of mathematics as time (Mathematics = Time). Contemporary theoretical physics and computer science are founded on Euler-Logic, a system characterised by static multiplication, squaring, and irrational root operations. This mathematical approach enforces rounding errors that manifest in the physical world as energetic resistance (entropy / waste heat). Current computing systems thereby reach an efficiency ceiling, while physical models merely simulate the law of conservation of energy while, in fact, violating it through logical friction.

Rather than artificially squaring dimensions, Fn-Logic follows the recursive addition rhythm of nature (fn_{n-1} + fn_{n-2} = fn_n). By dispensing with floating-point operations and returning to atomic integer addition, a system is created that is 100% consistent with the law of conservation of energy. The difference between the Euler prediction (122.625 m) and the Fn proof (127.53 m) after a fall time of 5 seconds in a vacuum marks the empirical divide between simulated and real physics.

Geometric Folding: Derivation of planetary density through recursive Fibonacci steps from equatorial circumference down to the atomic scale. Root Redundancy: Elimination of complex computational operations through the recognition that the information of a value's origin is already stored in the additive target value. IT Revolution: Design of a computer architecture that operates without FPUs (Floating Point Units), enabling lossless AI computations and graphics rendering at minimal thermal emission.

Future Vision: The implementation of Fn-Logic forms the foundation for a new era of terraforming and highly efficient energy conversion. By separating the atomic computational layer (back-end) from the human perception template (front-end), a technology is made possible that no longer works against nature but instead utilises its inherent rhythm.

Table of Contents

1. Foundations: The Mathematics of Process

1.1 Philosophy: Mathematics Is Time

In conventional Euler-Logic, numbers are regarded as static, lifeless values. Fn-Logic breaks with this tradition. Here, every number is the result of a continuous process of addition, in which the factor time governs the evolution of values.

> *Core statement: The universe does not calculate -- it adds. Every mathematical step is a real clock cycle of the processor (nature / atom).*

1.2 The Fundamental Process Chain

Fn-Logic eliminates the need for complex multiplications or exponentiations by using recursive addition. The formula structure is: fn_{n-1} + fn_{n-2} = fn_n, with initial values fn_0 = 1 and fn_1 = 1.

This produces an unbroken chain that natively represents the law of conservation of energy, since every new state is based on the complete sum of its predecessors.

Table 1-1: Fn Process Cycles and Parity (O-O-E Rhythm)

Step	Logic Process	Result	Parity
1	fn_0 + fn_1	fn_1	Odd (Impulse)
2	fn_1 + fn_1	fn_2	Even (Symmetry / Anchor)
3	fn_1 + fn_2	fn_3	Odd (Growth)
4	fn_2 + fn_3	fn_5	Odd (Tension)
5	fn_3 + fn_5	fn_8	Even (Structure / Resonance)

The rhythm O-O-E (Odd-Odd-Even) describes the constant alternation from asymmetry to symmetry. Time is the frequency at which this symmetry formation occurs.

Example: Fall Distance Calculation (5 seconds)

```
  9.81 + 9.81  m/s  =  19.62 m
 19.62 + 9.81  m/s  =  29.43 m
 29.43 + 19.62 m/s  =  49.05 m
 49.05 + 29.43 m/s  =  78.48 m
 78.48 + 49.05 m/s  = 127.53 m   <-- Fn fall distance

Euler: (127.53 / 13) x 12.5 = 122.625 m
Difference: 127.53 - 122.625 = 4.905 m  (= 9.81 / 2)
```

1.3 Why 12.5 (Euler) vs. 13 (Fn) Changes Everything

In Euler-Logic, time is squared and halved (0.5 * t^2). At 5 seconds (5^2 = 25), multiplying by 0.5 effectively means computing with the factor 12.5. In Fn-Logic, however, the 5th stage of the time-addition is inseparably linked to 13 (the next Fibonacci number after 8). 9.81 x 13 = 127.53 m.

1.4 The Refutation of Infinity

Fn-Logic postulates that the universe is atomic and finite. Since there are no half-atoms in nature, Fn-Logic uses only integers. Rounding is a mathematical error of Euler-Logic. In Fn-Logic, every calculation terminates upon reaching the atomic level. What we perceive as decimal numbers or rounding is merely a template laid over the vast, integral base number to make it measurable for human perception.

1.5 Energy Efficiency: Fn vs. Euler

While Euler-Logic generates up to 89% resistance (waste heat) through rounding errors and static models, Fn-Logic operates without loss. Euler calculates against the law of conservation of energy (friction loss through logical errors). Fn-Logic flows with the law of conservation of energy (100% conservation through additive process integrity).

2. Root Redundancy: The End of Root Mathematics

2.1 The Problem of Euler Squaring

In classical mathematics, dimensioning is often represented through x^2 (squaring). To return from this area back to the base unit (the vector), Euler-Logic requires the root operation (sqrt(x)). The deficiency: roots of non-perfect-square numbers yield irrational numbers (infinitely many decimal places). These infinite values do not exist in atomic reality. Computers consume enormous energy (waste heat) managing these rounding errors.

Table 2-1: Euler Path vs. Fn Path

Euler Path (Static)	**Fn Path (Processual)**	**Result**
2^2=4 -> sqrt(4)=2	1+1=2, 1+2=3, 2+3=5	No back-calculation needed
Produces area chaos	Produces linear time flow	Preservation of integer

2.2 Root Redundancy Through Time Integrity

Since every Fn state is the sum of its predecessors, the root (the origin of the value) is already stored in the current value.

> *Core principle: Whoever adds does not need to extract. In Fn-Logic, the root is not a computational step but the immediately preceding clock cycle of the system.*

2.3 Hardware Implications

A processor based on Fn-Logic requires no FPU (Floating Point Unit) for root calculations. Advantage: Massive reduction in transistor complexity. Effect: Computational operations at the speed of light with near-zero heat generation, since no irregular numbers need to be straightened out.

3. Conservation of Energy: Gravitation and Fn Integrity

3.1 The Euler Paradox: Apparent Conservation

In classical mechanics (Euler-Logic), the fall distance is calculated as s = 0.5 * g * t^2. At t = 5 s this yields approximately 122.625 m.

The problem of energy conservation: To arrive at this value, Euler-Logic must perform squarings (t^2) and roundings. Mathematically, this produces remainders (decimal places) that must be dissipated in physical reality as resistance heat (approx. 89% efficiency loss).

> *Critique: A system that requires waste heat to justify its computational errors violates the law of conservation of energy at the logical level.*

3.2 Fn-Logic: Real Conservation Through Addition

Fn-Logic does not calculate gravitation as a static curve but as an additive time increment. When time is treated as a 5-cycle processor (5 seconds) and acceleration is carried out as purely additive Fn steps (without rounding), the resulting fall distance is 127.53 m.

Why this represents true energy conservation:

1. Lossless translation: There are no irregular intermediate values. Every joule of potential energy is added precisely to kinetic energy (distance) in the Fn clock cycle.
2. 100% system integrity: Since Fn-Logic does not force any roots or squares, no energy needs to escape as heat.
3. Constancy of mass: Mass remains stable because the atomic integer logic leaves no room for mass defects from rounding errors.

3.3 The 5-Second Vacuum Check

The ultimate proof of the superiority of Fn-Logic over Euler-Logic takes place in a vacuum. Without air resistance, an object after 5 seconds must have reached exactly the Fn distance.

Model	Prediction (m)
Euler prediction	122.625 m (includes computational resistance)
Fn prediction	127.53 m (lossless energy addition)

3.4 Conclusion for Thermodynamics

Fn-Logic is the only mathematics that fulfils the law of conservation of energy 100%. While Euler consumes energy to compute, Fn-Logic is the energy itself.

> *The difference of approximately 5 metres between Euler and Fn is not a measurement error -- it is the proof of the logical friction of the old world.*

4. IT-Logic: The Integer Processor Without Floats

4.1 The Inefficiency of Current IT

Today's computer architectures are based on Euler-Logic and use Floating Point Units (FPUs) to compute with decimal numbers (floats). The problems: Rounding errors (computers cannot represent many decimal numbers exactly in binary). Energy loss from continuously managing and correcting rounding errors. Computational load: complex multiplications and divisions of floats consume millions of transistor cycles.

Table 4-1: Conventional IT vs. Fn-IT

Feature	Conventional IT (Float)	Fn-IT (Integer Clock)
Computation type	Complex multiply / divide	Pure, iterative addition
Precision	Limited by bits (rounding)	Absolute (atomically exact)
Heat	High (logical resistance)	Minimal (lossless flow)
Unit	Artificial 'floats'	Natural time cycles

4.2 The Template Interface (User Interface)

Fn-Logic strictly separates the computation layer from the perception layer. Back-end (core): The processor adds enormous integers without decimals. It operates losslessly and at the speed of light. Front-end (template): The operating system places a perception template over the result, scaling the large number to a human-readable magnitude (e.g., 127.53).

> *Important: The template does not change the number; it merely makes a part of the whole visible.*

4.3 Revolution for AI and Graphics

Artificial intelligence (AI) and graphics cards (GPUs) benefit the most from Fn-Logic. AI without weight rounding: Neural networks could perform trillions of operations without accumulating rounding errors -- an AI based on Fn would be 1000 times faster at the same power consumption. Real-time reality: Graphics are no longer approximated (ray-tracing simulation) but added atomically correctly.

4.4 Conclusion: Hardware-Level Relativity

By eliminating floats, we return to the simplicity of nature. An Fn processor is not a calculator but a time manipulator that maps reality in the rhythm of addition.

> *We must stop forcing the world into decimal numbers. We must start adding it.*

5. The Vacuum Drop Proof (Experiment)

5.1 Experimental Setup: The Dual-Tower Arrangement

To empirically demonstrate the discrepancy between Euler-Logic and Fn-Logic, an experiment is proposed using two identical vacuum drop tubes. Test objects: Two identical lead spheres (minimal surface friction, maximum density). Medium: High vacuum (exclusion of air resistance).

Tower A (Euler reference): Drop height 122.625 m. Tower B (Fn reference): Drop height 127.53 m.

5.2 The Prediction Matrix (Extended)

The core of the experiment lies in the discrepancy between the static approximation (Euler) and the processual reality (Fn). The following matrix shows how time cycling determines the physical distance.

System	Drop Height (m)	Expected Time (s	Factor	Logic Status
Euler Model	122.625 m	5.000 s	12.5	Static curve (erroneous)
Fn-Logic	122.625 m	4.500 s	12.5	Accelerated clock advantage
Fn-Logic	127.530 m	5.000 s	13.0	Real symmetry (nature)
Euler Model	127.530 m	5.099 s	13.0	Logical friction (delay)

5.3 Analysis of the Matrix Values

The 0.5-Jump (Time Advantage): In Fn-Logic, the object reaches the Euler mark (122.625 m) after only 4.5 seconds. Classical physics requires an additional 0.5 seconds of buffer to conceal its mathematical imprecision. This buffer is the root of the efficiency loss in all Euler-based systems.

The Factor Difference (12.5 vs. 13): Euler rigidly computes with the factor 12.5 (0.5 x 25). This corresponds to a half logic that slows the energy flow. Fn-Logic uses the natural factor 13 (the 5th stage of Fn addition). The 13 is the integer resonance of nature. Result: 9.81 x 13 = 127.53 m.

5.4 Conclusion

The vacuum drop experiment represents the empirical falsification of the Euler model. The Fn-Logic prediction of 127.53 m after exactly 5 seconds reflects the lossless nature of integer addition. The Euler delay -- the 0.5-second discrepancy -- is not a measurement artefact but the measurable signature of logical friction inherent to floating-point mathematics.

References & Citations

This work references the following prior works by the author:

[1] Tensor Realities (TdR)

Working Paper

DOI: https://doi.org/10.5281/zenodo.18727529

GitHub: RaikoPulvermacher/Tensor-Realit-ten @ v1.04

[2] Energy Revolution

Preprint

DOI: https://doi.org/10.5281/zenodo.18778354

GitHub: RaikoPulvermacher/Energie-Revolution- @ v1.0

Author Contact

Author: Raiko Pulvermacher

E-Mail: Pulvermacher.Raiko@web.de

ORCID ID: 0009-0003-9431-1001 (https://orcid.org/0009-0003-9431-1001)

Open Science Framework (OSF): https://osf.io/py42t/

X/Twitter: https://x.com/RaPulvermacher

TikTok: https://www.tiktok.com/@raiko.pulvermacher

License: Pulvermacher Open Research License (PORL) v1.0

DOI: https://doi.org/10.5281/zenodo.18757232

License Repository: https://github.com/RaikoPulvermacher/PORL/tree/v.1.0

Deutsch / German

Diese Inhalte (Texte, Bilder, Konzepte) duerfen genutzt, geteilt und weiterentwickelt werden unter den folgenden Bedingungen:

1. Namensnennung
Bei jeder Nutzung, Weitergabe, Veroeffentlichung oder Zitierung muss der Urheber klar und sichtbar genannt werden: Raiko Pulvermacher

2. Transparenz bei Weiterentwicklung
Wenn Inhalte veraendert, erweitert oder in neue Kontexte eingebettet werden, muss klar kenntlich gemacht werden, dass es sich um eine abgeleitete Arbeit handelt.

3. Forschung & Publikationen
Wird diese Arbeit als Grundlage fuer wissenschaftliche Untersuchungen, Experimente, Simulationen, Veroeffentlichungen, Preprints oder Vortraege verwendet, ist der Urheber vorab oder spaetestens bei Veroeffentlichung zu informieren. Eine Mitautorenschaft wird erwartet, wenn diese Arbeit inhaltlich wesentlich beitraegt.

4. Keine Ausschliesslichkeit
Diese Lizenz erhebt keinen Anspruch auf Kontrolle ueber Ergebnisse, sondern auf Transparenz, Anerkennung und Beteiligung.

5. Haftungsausschluss
Diese Arbeit stellt keine physikalische Theorie, kein formales Modell und keine ueberprufte wissenschaftliche Aussage dar. Sie wird ohne Garantie bereitgestellt. Die Nutzung erfolgt auf eigenes Risiko.

6. Lizenzweitergabe
Abgeleitete Arbeiten muessen diese Lizenz oder eine inhaltlich gleichwertige Lizenz beibehalten.

English / Englisch

These contents (texts, images, concepts) may be used, shared, and further developed under the following conditions:

Contact:

Raiko Pulvermacher

E-Mail: Pulvermacher.Raiko@web.de

ORCID: https://orcid.org/0009-0003-9431-1001

OSF: https://osf.io/py42t/